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CLINICAL RESEARCH

The Accuracy of Initial U.S. Heart Transplant Candidate Rankings

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ABSTRACT

BACKGROUND The U.S. heart allocation system ranks candidates with only 6 treatment-based categorical "statuses" and ignores many objective patient characteristics.

OBJECTIVES This study sought to determine the effectiveness of the standard 6-status ranking system and several novel prediction models in identifying the most urgent heart transplant candidates.

METHODS The primary outcome was death before receipt of a heart transplant. The accuracy of the 6-status system was evaluated using Harrell's C-index and log-rank tests of Kaplan-Meier estimated survival by status for candidates listed postpolicy (November 2018 to March 2020) in the Scientific Registry of Transplant Recipients data set. The authors then developed Cox proportional hazards models and random survival forest models using prepolicy data (2010-2017). The predictor variables included age, diagnosis, laboratory measurements, hemodynamics, and supportive treatment at the time of listing. The performance of these models was compared with the candidate's 6-status ranking in the postpolicy data.

RESULTS Since policy implementation, the 6-status ranking at listing has had moderate ability to rank-order candidates (C-index: 0.67). Statuses 4 and 6 had no significant difference in survival ($P = 0.80$), and status 5 had lower survival than status 4 ($P < 0.001$). Novel multivariable prediction models derived with prepolicy data ranked candidates correctly more often than the 6-status rankings (Cox proportional hazards model C-index: 0.76; random survival forest model C-index: 0.74). Objective physiologic measurements, such as glomerular filtration rate, had high variable importance.

CONCLUSIONS The treatment-based 6-status heart allocation system has only moderate ability to rank-order candidates by medical urgency. Predictive models that incorporate physiologic measurements can more effectively rank-order heart transplant candidates by urgency. (J Am Coll Cardiol HF 2023;11:504–512) © 2023 Published by Elsevier on behalf of the American College of Cardiology Foundation.

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Heart transplantation is the definitive treatment for end-stage heart failure and is associated with a large improvement in 5-year survival.[1] Despite a steady increase in heart transplants over the past decade, the demand for donor hearts far exceeds supply, and over one-third of candidates die or are delisted without a transplant.[2] In contrast with other solid organs, the Organ Procurement and Transplant Network ranks heart transplantation candidates with categorical "statuses" that are based primarily on the intensity of supportive treatment.[3] This system ignores many objective patient characteristics, such as glomerular filtration rate (GFR), that are strongly associated with the risk of death without heart transplantation.[4]

In October 2018, the heart allocation policy was revised to identify more accurately and treat, through heart transplantation, the most urgent candidates, thus expanding the geographic sharing of donor hearts and increasing the number of statuses from 3 to 6. Although the 6-status policy incorporates some limited physiologic data, it still has few risk categories and relies heavily on treatment as a proxy for medical urgency. Multivariable prediction models may be significantly more accurate than the 6-status system and may better fulfill the federal requirement to rank-order candidates by urgency.[4]

This observational registry cohort study aimed to determine the discrimination (rank-ordering accuracy) of the therapy-based 6-status heart allocation system compared with novel multivariable prediction models. We hypothesized that both conventional (Cox proportional hazards [CPH]) and more sophisticated machine learning (random survival forest [RSF]) survival analysis techniques would generate more accurate heart transplant candidate rankings than the 6-status system in a cohort of candidates listed under the new allocation policy.

METHODS

DATA SOURCE AND OUTCOME. This retrospective study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donors, wait-listed candidates, and transplant recipients in the United States, submitted by the members of the Organ Procurement and Transplantation Network. The Health Resources and Services Administration of the U.S. Department of Health and Human Services provides oversight for the activities of the network and registry contractors. This study was declared exempt from review by the University of Chicago Institutional Review Board.

The study group consisted of all adult candidates (at least 18 years of age) listed for heart transplantation (excluding heart-lung candidates). The training set for the multivariable prediction models consisted of individuals listed between January 1, 2010 and December 31, 2017, thereby spanning the prepolicy period in the era of modern continuous flow left ventricular assist devices (LVADs).[5] The test set consisted of individuals listed between November 1, 2018, and March 1, 2020, the period after the new heart allocation policy was implemented. Candidates who had values of height <30 cm, body mass index (BMI) >100 kg/m^2, creatinine <0.1 mg/dL, missing 6 status, or a death date preceding the listing date were excluded as data entry errors.

The primary outcome of the study was death without receiving a heart transplant, including death while on the waitlist and after delisting (as recorded in the Social Security Death Master File). Heart transplantation was treated as a censoring (not competing) event because the goal was to rank-order candidates correctly where transplantation was not an option and identify the effects of covariates on the cause-specific hazard function.[6] Candidates who were delisted for reasons other than transplantation and did not die during the study period were censored at delisting.

PREDICTOR VARIABLES. Variables were divided into 7 conceptual categories for joint variable importance tests: physiologic measurements, treatments, cardiac diagnoses, candidate blood type, medical history, and other. Physiologic measurements included estimated GFR,[7] pulmonary capillary wedge pressure (PCWP), mean pulmonary artery pressure (mean PAP), and cardiac index (calculated from cardiac output and Mosteller's formula for body surface area).[8] Treatments included intra-aortic balloon pumps (IABPs), intravenous (IV) inotropes, extracorporeal membrane oxygenation (ECMO), LVADs, and other ventricular assist devices (VADs). We grouped cardiac diagnoses into 7 categories: nonischemic dilated cardiomyopathy; ischemic cardiomyopathy; congenital heart disease; restrictive cardiomyopathy; hypertrophic cardiomyopathy; valvular heart disease; and other. Candidate blood type was given by the variables A, AB, B, or O. Medical history included a history of malignant disease, cerebrovascular disease, and tobacco use. Age, BMI, and diabetes were assigned to an "other variables" category. To test for differences between the training and test cohorts, we used

ABBREVIATIONS AND ACRONYMS

BMI = body mass index
CPH = Cox proportional hazard
ECMO = extracorporeal membrane oxygenation
GFR = glomerular filtration rate
IABP = intra-aortic balloon pump
IV = intravenous
LVAD = left ventricular assist device
PAP = pulmonary artery pressure
PCWP = pulmonary capillary wedge pressure
RSF = random survival forest
VAD = ventricular assist device

Wilcoxon rank sum tests for continuous variables and chi-square tests for categorical variables. The listing year was also included as a predictor variable in all calculations.

For indicator variables, missing data were assumed to represent the absence of the condition. Other missing data were imputed with the multiple imputation by chained equations algorithm by using the mice package in R software (R Foundation).[9]

STATISTICAL METHODS. We used 2 multivariable modeling techniques. First, we fit a standard unpenalized Cox proportional hazards (CPH) model.[10,11] Second, we used a random survival forest (RSF) algorithm using the randomForestSRC package in R software.[12,13] RSF applies the basic principles of random forests to the problem of survival, thus assigning a cumulative hazard function to each terminal node. We fitted models with and without treatment variables as input features for each technique, resulting in 4 total models. Variable importance was calculated for the RSF models by using the "anti" method, on the basis of the measurement of prediction error obtained when dropping out-of-bag cases into the trained decision trees. With joint variable importance, the change in prediction error is computed when multiple variables are simultaneously assigned to the "anti" switching behavior. We calculated joint variable importance for each category of predictor variable described earlier.

We assessed the performance of the 6-status system and the novel multivariable prediction models by using several different metrics. Harrell's C-index is a concordance index that considers pairs of individuals and measures the model's goodness of fit in predicting their relative risk.[14] Differences between C-indices were assessed in 2 ways. First, we calculated a *P* value from a Wald test of the difference between C-indices with 2-sided hypothesis testing by using bootstrapping to obtain the SD of the difference. Second, we calculated the rank correlation of paired predictions, defined as the fraction of relevant pairs for which 1 method is concordant (correct in predicting the relative risk of a pair) and the other method is not concordant.[15] We calculated 95% CIs for the C-indices by using the boot package in R software.[16,17]

To assess the calibration of the 6-status system, we estimated Kaplan-Meier survival curves for each status category in the test data set. To assess the calibration of our novel multivariable prediction models, we divided our test cohort into 4 clinically meaningful groups by predicted 90-day survival probability. These were defined as follows: group 1, <85%; group 2, 85%-90%; group 3, 90%-95%; and group 4, ≥95%. We estimated Kaplan-Meier survival functions for each group, and the value of the Kaplan-Meier survival function at 90 days was compared among the groups. Pairwise log-rank tests were used to test for the significance of differences among adjacent survival curves up to 365 days. We also calculated Brier scores for both models.[18,19]

Finally, to examine the relationship between the 6-status rankings and the model-predicted survival probabilities, we created box plots showing the median and IQR of model-predicted 90-day survival probability for each of the 6-status groups. See the Supplemental Methods for more details on the statistical approach.

RESULTS

STUDY GROUP. A total of 33,309 adult heart transplant candidates were listed in the study periods. We excluded 418 heart-lung candidates, 483 patients with data entry errors, and 114 patients who were listed as inactive. The final data set contained 32,294 adult heart transplant candidates (mean age: 53.0 years; 73.7% male), 27,200 candidates in the prepolicy training set and 5,094 candidates in the postpolicy test set (Supplemental Figure 1). Candidate characteristics in both cohorts are shown in **Table 1**. Use of IABP at initial listing increased from 4.7% prepolicy to 13.2% postpolicy. ECMO use increased from 1.6% prepolicy to 2.8% postpolicy. The percentage of cases with missing data requiring imputation was 0.3% for GFR, 6.4% for cardiac index, 4.8% for mean PAP, 9.7% for PCWP, and 0.3% for BMI.

RANK-ORDERING PREDICTIVE PERFORMANCE. After implementation with the postpolicy cohort, the 6-status rankings at listing showed moderate rank-ordering ability (Harrell's C-index [c_H] = 0.67; 95% CI: 0.63-0.71). Both the CPH with treatment (CPH-T) and RSF with treatment (RSF-T) models better discriminated between lower-risk and higher-risk candidates compared with the 6-status system (CPH-T: c_H = 0.76 [95% CI: 0.72-0.79]; $P < 0.001$; RSF-T: c_H = 0.74 [95% CI: 0.70-0.78]; $P = 0.011$) (**Table 2**). For 30.2% of the relevant pairs in the postpolicy cohort, CPH-T was concordant and the 6-status rankings were not concordant. For 17.2%, the 6-status rankings were concordant and CPH-T was not. For 30.5%, RSF-T was concordant and the 6-status rankings were not. For 19.7% of pairs, the 6-status rankings were concordant and RSF-T was not. Eliminating treatment variables (denoted CPH-N and RSF-N) reduced the predictive models' C-indices

to levels that were only slightly higher than the 6-status system (CPH-N: c_H = 0.69 [95% CI: 0.66-0.73]; RSF-N: c_H = 0.70 [95% CI: 0.67-0.74]). For 30.8%, CPH-N was concordant and 6-status rankings were not. For 24.8%, the 6-status rankings were concordant and CPH-N was not. For 30.2%, RSF-N was concordant and the 6-status rankings were not. For 23.7%, the 6-status rankings were concordant and RSF-N was not.

At 90 days, the Brier score was 0.0340 for CPH-T, 0.0345 for RSF-T, 0.0366 for CPH-N, and 0.0379 for RSF-N. Supplemental Figure 2 shows Brier scores for the 4 predictive models for 365 days post listing.

VARIABLE IMPORTANCE. Figures 1A and 1B show individual variable importance in the RSF models. With RSF-T, GFR has the highest importance at 0.093, followed by ECMO at 0.089. Other variables with relatively high importance were age (0.035), PCWP (0.026), other VAD (0.025), IABP (0.024), mean PAP (0.024), IV inotropes (0.024), and cardiac index (0.023). The remaining 16 variables all had an importance of <0.02 in RSF-T. The variable importance scores for RSF-N were similar. The variables with importance >0.02 were GFR (0.081), age (0.038), PCWP (0.036), valvular heart disease (0.025), mean PAP (0.024), and cardiac index (0.023). The magnitude of the HRs of the individual covariates in the CPH were consistent with the RSF variable importance (Supplemental Tables 1 and 2). ECMO was associated with a 297% increase in risk (HR: 3.97 [95% CI: 3.4-4.6]), and an increase in GFR from 50 to 100 mL/min was associated with a 40% reduction in risk (HR: 0.990 for a 1-unit change in GFR [95% CI: 0.989-0.991]).

Figures 1C and 1D show joint variable importance by category. In RSF-T, the category of physiologic measurements ranked highest, with a joint importance of 0.113. The treatment category ranked second, with an importance of 0.083. The "other" category had an importance of 0.035, and the remaining categories (heart failure diagnosis, blood type, and history) had a joint importance of <0.01. In RSF-N, the category of physiologic measurements had a joint importance of 0.153, over 3 times the value of the second-ranking category of "other" variables (0.039). All other categories had a joint importance of <0.02.

CALIBRATION OF THE PREVIOUS 6-STATUS CLASSIFICATION SYSTEM AND THE NOVEL MODELS. Both the CPH-predicted 90-day survival risk groups and the RSF-predicted 90-day survival risk groups had the expected ordering, with lower-risk groups having significantly greater survival probability than higher-risk groups (Figure 2, Supplemental Figures 3 to 5).

TABLE 1 Baseline Characteristics of Transplant Candidates

	Prepolicy (n = 27,200)	Postpolicy (n = 5,094)	*P* Value
Physiologic measurements			
GFR, mL/min	62.2 (47.1-79.4)	61.8 (46.7-78.8)	0.115
PCWP, mm Hg	19 (13-25)	18 (11-24)	<0.001[a]
Pulmonary artery mean pressure, mm Hg	29 (21-36)	27 (20-35)	<0.001[a]
Cardiac index, L/min/m^2	2.09 (1.7-2.5)	2.07 (1.7-2.5)	0.011[b]
Treatments			
IABP	1,279 (4.7)	674 (13.2)	<0.001[a]
IV inotropes	8,550 (31.4)	1,539 (30.2)	0.087
Extracorporeal membrane oxygenation	422 (1.6)	143 (2.8)	<0.001[a]
LVAD	7,184 (26.4)	1,331 (26.1)	0.901
Other ventricular assist device	971 (3.6)	180 (3.5)	
Heart failure diagnosis			<0.001[a]
Nonischemic dilated cardiomyopathy	13,891 (51.1)	2,748 (53.9)	
Ischemic cardiomyopathy	9,250 (34.0)	1,429 (28.1)	
Congenital heart disease	907 (3.3)	259 (5.1)	
Restrictive cardiomyopathy	882 (3.2)	221 (4.3)	
Hypertrophic cardiomyopathy	588 (2.2)	156 (3.1)	
Valvular heart disease	382 (1.4)	53 (1.0)	
Other	1,300 (4.8)	228 (4.5)	
Blood type			0.321
A	10,211 (37.5)	1,900 (37.3)	
AB	1,318 (4.8)	236 (4.6)	
B	3,772 (13.9)	755 (14.8)	
O	11,899 (43.7)	2,203 (43.2)	
History			
Malignant disease	2,249 (8.3)	465 (9.1)	0.045[b]
Cerebrovascular disease	16,01 (5.9)	352 (6.9)	0.005[b]
Tobacco use	12,967 (47.7)	2,152 (42.2)	<0.001
Other variables			
Age, y	56 (46-63)	56 (46-63)	0.492
BMI, kg/m^2	27 (24-31)	28 (24-32)	<0.001[a]
Diabetes	8,133 (29.9)	1,517 (29.8)	0.876

Values are median (IQR, 25th-75th percentile) or n (%). *P* values are shown for Wilcoxon rank sum statistic (continuous variables) and chi-square test (categorical variables) that test for differences between the cohorts. [a]$P < 0.001$. [b]$P < 0.010$.

BMI = body mass index; GFR = glomerular filtration rate; IABP = intra-aortic balloon pump; IV = intravenous; LVAD = left ventricular assist device; PCWP = pulmonary capillary wedge pressure.

For CPH-T, all adjacent curves were significantly different at $P < 0.001$. For RSF-T, the group 1/group 2 log-rank test was significantly different at $P < 0.001$, and the group 2/group 3 and group 3/group 4 tests

TABLE 2 Harrell's C-Indices, Tested on the Postpolicy Cohort

	With Treatment Variables	With No Treatment Variables
Cox proportional hazards	0.76 (0.72-0.79)[a]	0.69 (0.66-0.73)
Random survival forest	0.74 (0.70-0.78)[b]	0.70 (0.67-0.74)
6-status rankings	0.67 (0.63-0.71)	

Values are Harrell's C-index (95% CI). CIs were calculated with the normal approximation to bootstrapped results, using 250 bootstrapped replications. A higher C-index indicates better discrimination between lower-risk and higher-risk candidates. Footnotes reflect the *P* value of the Wald test comparing the model-based C-index with the 6-status C-index. [a]$P < 0.001$. [b]$P < 0.050$.

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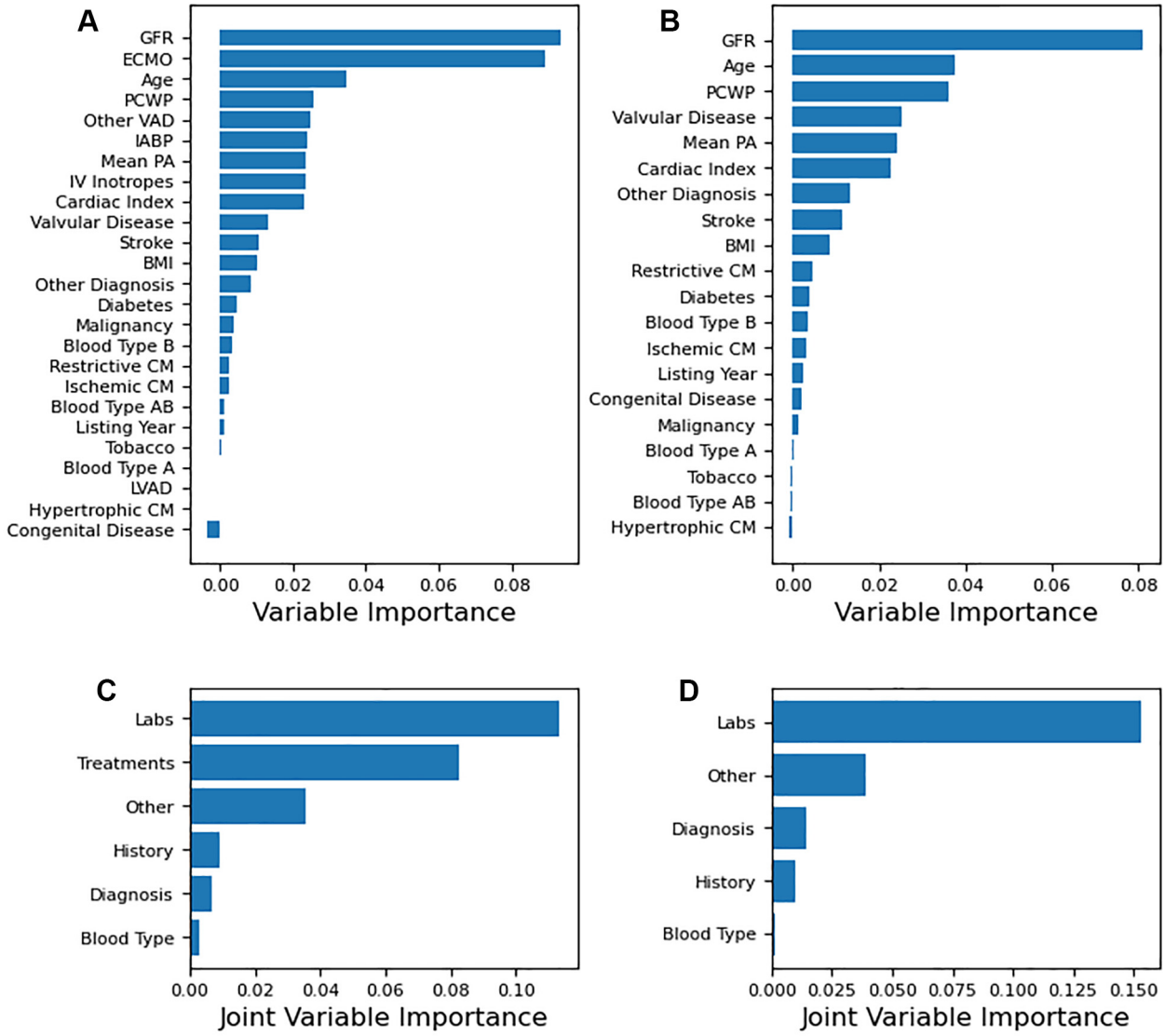
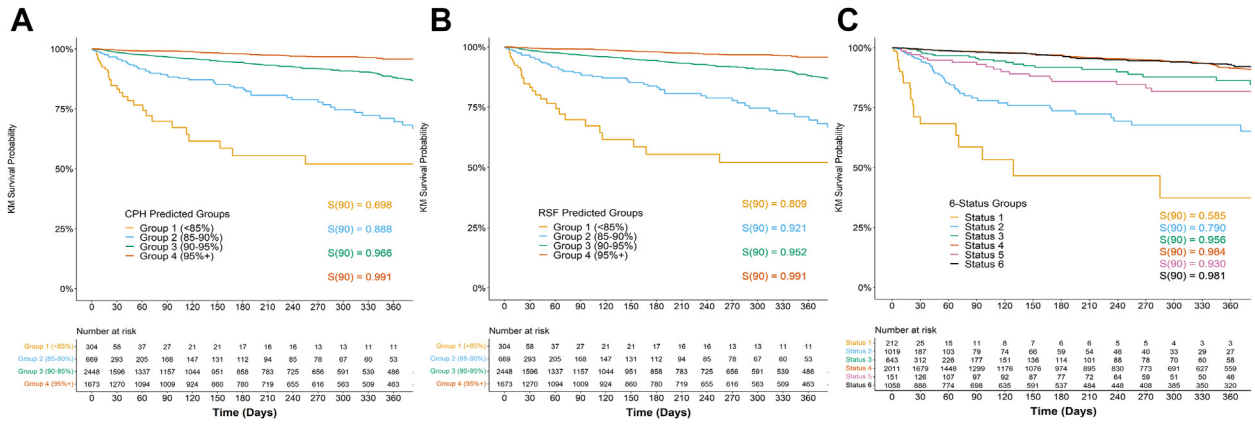
FIGURE 1 Variable Importance in the Random Survival Forest Models



Single variable (KM) importance for calculations that **(A)** include treatment variables and **(B)** exclude treatment variables. Joint variable importance for calculations that **(C)** include treatment variables and **(D)** exclude treatment variables. Laboratory (Labs) and physiologic variables include glomerular filtration rate (GFR), pulmonary capillary wedge pressure (PCWP), pulmonary artery (PA) mean pressure, and cardiac index. Treatment variables include intra-aortic balloon pump (IABP), intravenous (IV) inotropes, extracorporeal membrane oxygenation (ECMO), left ventricular assist device (LVAD), and other ventricular assist device (VAD). Other variables include age, body mass index (BMI), and diabetes. History variables include malignancy, cerebrovascular disease, and tobacco use. Diagnosis variables include nonischemic dilated cardiomyopathy (CM), ischemic cardiomyopathy, congenital heart disease, restrictive cardiomyopathy, hypertrophic cardiomyopathy, valvular heart disease, and other diagnoses.

were significant at $P = 0.004$ and $P = 0.002$, respectively. For the 6-status rankings, statuses 1 to 3 had significant differences in survival at $P < 0.001$, and statuses 3 and 4 differed with $P = 0.002$. However, among statuses 4 to 6, overlapping of curves was observed, as well as incorrect ordering of nonoverlapping survival curves. Status 5 candidates had significantly worse survival than status 4 candidates ($P < 0.001$) and the survival of status 4 and 6 candidates was not significantly different ($P = 0.80$). The

FIGURE 2 Kaplan-Meier Survival by Groups From the CPH, RSF, and 6-Status Models



Kaplan-Meier (KM) survival and risk tables for **(A)** groups on the basis of predicted 90-day survival from the Cox proportional hazards (CPH) model, **(B)** the same groups on the basis of predicted 90-day survival from the random survival forest (RSF) model, and **(C)** groups on the basis of statuses from the 6-status ranking. For **(A)** and **(B),** the groups are group 1 (<85% 90-day survival), group 2 (85%-90%), group 3 (90%-95%), and group 4 (≥95%). S(90) shows the Kaplan-Meier predicted 90-day survival.

Kaplan-Meier estimated 90-day survival was 59% for status 1, 79% for status 2, 96% for status 3, 98% for status 4, 93% for status 5, and 98% for status 6.

MODEL-PREDICTED SURVIVAL OF STATUS GROUPS. The **Central Illustration** shows the predicted 90-day survival by 6-status ranking group for CPH-T and RSF-T. Status 1 candidates had the lowest predicted survival in both models, with a median predicted survival of 83.2% (IQR: 71.4%-92.6%) (CPH-T) and 68.2% (IQR: 53.7%-85.9%) (RSF-T). Both models showed similar predicted survival for statuses 4 and 6, contrasted with lower survival for status 5. CPH-T predicted median survival of 94.4% for status 4, 91.1% for status 5, and 94.6% for status 6. RSF-T showed median survival of 95.7% for status 4, 89.4% for status 5, and 95.9% for status 6.

DISCUSSION

In this registry cohort study of 32,294 heart transplant candidates, we found that the 6-status heart allocation system had limited accuracy. Only candidates listed in statuses 1 to 4 had significant differences in survival, and status 5 candidates were not correctly ordered. Both the RSF and CPH predictive models had higher Harrell's C-indices relative to the 6-status system. Objective physiologic measurements, GFR in particular, had high variable importance for predicting waitlist mortality.

A strength of the current 6-status system is that it can identify 3 high-risk groups that have significantly different survival curves (statuses 1, 2, and 3). These results are consistent with the finding that waitlist survival has improved overall since the policy change,[20-22] which divided the overcrowded status 1a group into statuses 1 to 3.[2] Paired with the finding that post-transplant survival has not significantly changed,[21,23,24] this makes the 6-status system unequivocally an improvement. However, our work clearly shows that discriminative power increases when a broader set of characteristics, including physiologic measurements, is considered.

The joint variable importance scores show that objective physiologic measurements have a powerful effect on the discrimination of predictive models. GFR demonstrated high single variable importance, offering a minimally invasive physiologic measurement with significant predictive power in assessing medical urgency. Because most status 5 candidates are in renal failure (listed for heart-kidney transplantation without any other treatments to qualify for higher status), this may explain the lower survival in this patient group. These results suggest that the inclusion of more objective physiologic data (including data that would be easily measured by blood tests) could improve the accuracy of the heart allocation system in identifying and prioritizing the most medically urgent candidates. Our single variable important scores are consistent with the findings of Hsich et al,[4] who applied RSF to an earlier cohort of adult heart transplant candidates listed from 2004 to 2015. Their work, which evaluated variable

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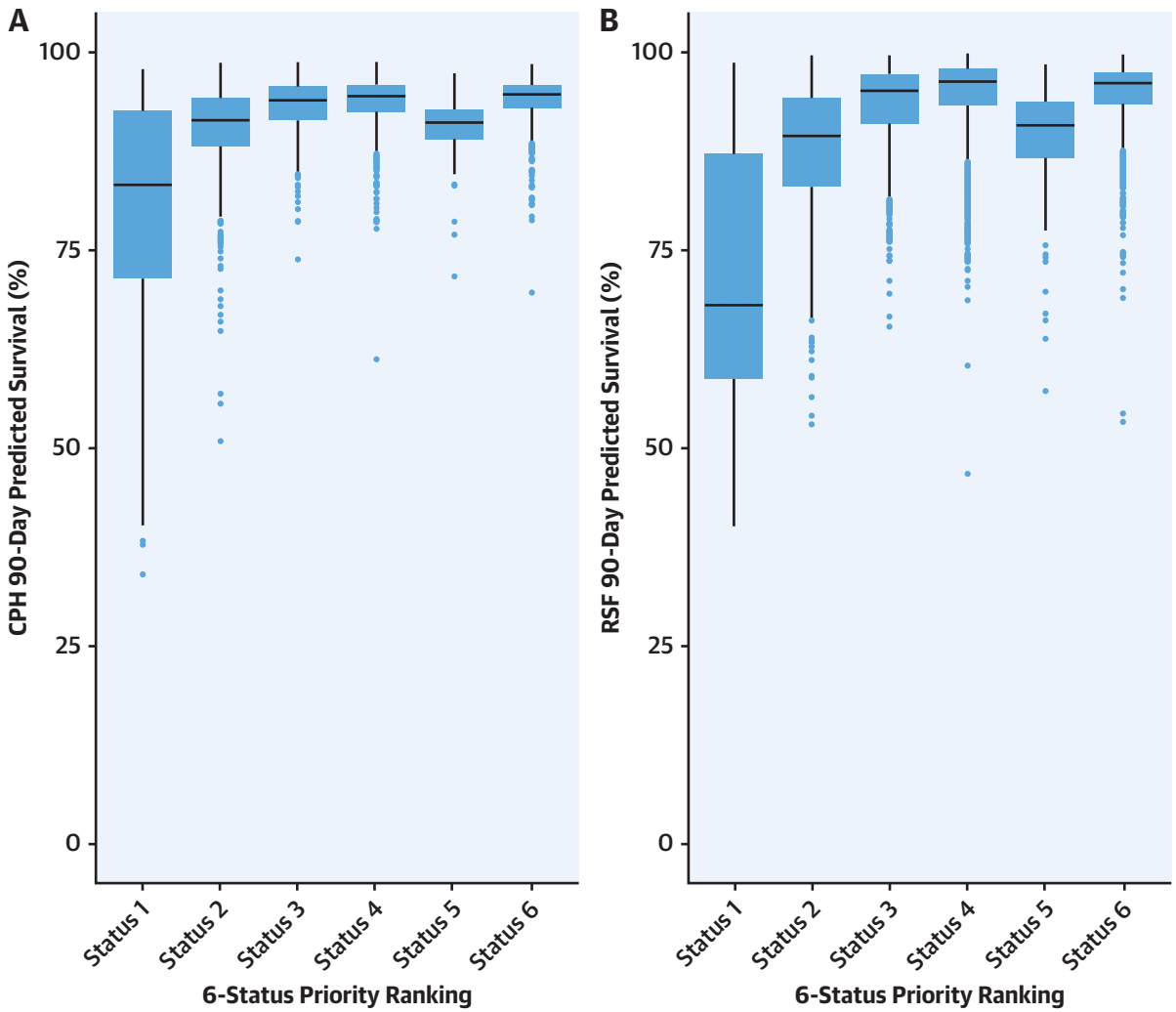
CENTRAL ILLUSTRATION **Predicted 90-Day Survival by 6-Status Ranking**



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Box plots showing median and IQR range (25th, 75th percentile) for the predicted 90-day survival probabilities from the **(A)** Cox proportional hazards (CPH) model with treatment and the **(B)** random survival forest (RSF) model with treatment.

importance by the permutation method, identified GFR as the most important variable in predicting survival. In fact, we demonstrate that prediction models that are based only on nontreatment data can perform just as well as the 6-status system. Finally, the heterogeneity of predicted risk within status 1 and status 2 candidates suggests that the current cardiogenic shock criteria are not sufficient to ensure that only the most urgent candidates are listed at these statuses.

However, although physiologic measurements are more objective than treatment decisions, we should not assume that a heart allocation score incorporating objective physiologic measurements will be resistant to manipulation. For example, hemodynamic variables could be deliberately altered by measurement technique, timing, or concomitant drug administration. Detecting inappropriate manipulation of patient variables, whether in treatments or physiologic measurements, may be achievable with case-by-case study by a medical expert, but it presents a challenge for conventional statistical methods. Although the work presented here shows similar discriminative power for CPH and RSF, we note that RSF offers an

advantage in its ability to learn significant interactions from the data, which may succeed in detecting such manipulation. Capturing this expert intelligence in a computational algorithm may be improved by other machine learning techniques such as neural networks.

Overall, our results suggest that a heart allocation policy that is based on predictive modeling would better serve the U.S. Department of Health and Human Services final rule: to define an allocation policy that makes the "best use" of scarce donor hearts by ranking candidates from "most to least urgent."[25] Basing allocation primarily on subjective treatment decisions is not only inefficient but also inequitable. Previous work demonstrates that some heart transplant centers overtreat candidates with intensive therapies at far higher rates than other centers,[26-28] and they had widely varying increases in the use of IABP and ECMO and transplantation rates after implementation of the 6-status system.[21,29-32] These differences in treatment practices disadvantage candidates listed at centers that reserve intensive therapy for the sickest candidates and subject less urgent patients to unnecessary invasive procedures.

STUDY LIMITATIONS. The specific models presented here are not intended to be used directly as heart allocation scores but rather to demonstrate the disadvantages of a treatment-based system with only 6 categories relative to a continuous risk score incorporating a broader range of characteristics. More complex multivariable prediction models that account for time-dependent covariates and use all data gathered from listing to the time of death, transplantation, or delisting are needed.

Another limitation is the inability to distinguish between waitlist mortality caused by disease and mortality caused by complications of an intervention. Our results showed that ECMO, IABP, IV inotropes, and other VADs have positive HRs at $P < 0.001$ and relatively high variable importance. However, both HRs and variable importance may be artificially inflated by cases in which death was caused by treatment complications (a distinction that is not recorded in this data set). This issue underscores the importance of considering a range of objective, non-treatment-based variables in ranking the urgency of heart transplant candidates.

CONCLUSIONS

This work demonstrates that both conventional CPH analysis and RSF machine learning models can outperform the 6-status ranking at listing in discriminating candidates by medical urgency. We find that objective physiologic measurements substantially improve the prediction of waitlist mortality.

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The data reported here have been supplied by the Hennepin Healthcare Research Institute as the contractor for the Scientific Registry of Transplant Recipients (SRTR). The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy of or interpretation by the SRTR or the U.S. Government. Dr Parker has received funding from National Institutes of Health grant K08HL150291. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: The 6-status ranking system for heart transplant allocation had limited ability in selecting candidates on the basis of medical urgency. CPH and RSF models that incorporated physiologic measurements were able to generate better rankings.

TRANSLATIONAL OUTLOOK: Using predictive models that incorporate a wide range of patient characteristics could lead to a more efficient and equitable distribution of scarce donor organs.

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APPENDIX For an expanded Methods section as well as supplemental figures and tables, please see the online version of this paper.